

6.3) $D_f = [1, 5]$

skalární sklon dle vzorce $\langle f, g \rangle = \int_1^5 f(x)g(x)dx$

a) Ortonormalizační báze podprostoru $\text{Span}\{1, x, x^2\}$ prostoru V pomocí Gram-Schmidtovy metody:

$$1) \vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} \cdot 1 = \frac{1}{\sqrt{4}}$$

$$\vec{u}_2 = \vec{v}_2 - \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1$$

$$\vec{u}_2 = x - \left(\int_1^5 x dx \right) \frac{1}{\sqrt{4}} = x - \frac{1}{2} = \frac{x-1}{2} \quad \vec{v}_2 = \frac{x-3}{\sqrt{4}} = \frac{x-3}{2} = \frac{\sqrt{3}}{4} (x-3)$$

$$\int_1^5 x dx = \frac{1}{2} x^2 \Big|_1^5 = \frac{1}{2} \left(\frac{25}{2} - \frac{1}{2} \right) = \frac{1}{2} \cdot \frac{24}{2} = \frac{6}{2} = 3$$

$$\left[\frac{(x-3)^2}{2} \right]_1^5 = \frac{1}{2} - \left(-\frac{8}{2} \right) = \frac{9}{2}$$

$$\vec{u}_3 = \vec{v}_3 - \left(\langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2 \right)$$

$$\vec{u}_3 = x^2 - \left(\int_1^5 x^2 dx \right) \frac{1}{\sqrt{4}} + \left(\int_1^5 (x-3)x dx \right) \frac{\sqrt{3}}{4} (x-3)$$

$$= x^2 - \left(\int_1^5 x^2 dx \right) \frac{1}{2} + \left(\int_1^5 (x^2 - 3x) dx \right) \frac{\sqrt{3}}{4} (x-3)$$

$$\int_1^5 x^2 dx = \frac{1}{3} x^3 \Big|_1^5 = \frac{1}{3} \left(\frac{125}{3} - \frac{1}{3} \right) = \frac{1}{3} \cdot \frac{124}{3} = \frac{124}{9}$$

$$\left(\int_1^5 (x^2 - 3x) dx \right) \frac{\sqrt{3}}{4} (x-3) = \left(\frac{1}{3} \int_1^5 (x^3 - 9x^2) dx \right) \frac{\sqrt{3}}{4} (x-3)$$

$$\left(\frac{1}{3} \left(\frac{x^4}{4} - 9 \cdot \frac{x^3}{3} \right) \right) \frac{\sqrt{3}}{4} (x-3) = \frac{1}{12} (x^4 - 9x^3) \frac{\sqrt{3}}{4} (x-3)$$

$$\frac{1}{12} \cdot 32 \cdot (x-3)$$

$$\vec{u}_3 = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{238514}}$$

ON báze podprostoru je $u_1 = \frac{1}{2}$; $u_2 = \frac{\sqrt{3}}{4}(x-3)$; $u_3 = x^2 - 6x + \frac{23}{3}$

b) O6 projekce $\sin(x)$ na $\text{Span}\{1, x, x^2\}$ vyjádření jako $ax^2 + bx + c$

$\text{proj}(\sin(x)) = ax^2 + bx + c$

$$P(x) = \langle u_1, \sin(x) \rangle u_1 + \langle u_2, \sin(x) \rangle u_2 + \langle u_3, \sin(x) \rangle u_3$$

$$\langle u_1, \sin(x) \rangle = \int_1^5 \sin(x) dx = \int_1^5 \sin(x) dx = 0,12832 \rightarrow (0,12832) \frac{1}{2} = 0,06416$$

$$\langle u_2, \sin(x) \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin(x) dx = \frac{\sqrt{3}}{4} \int_1^5 (x-3) \sin(x) dx = -1,4932 \rightarrow (-1,4932) \frac{\sqrt{3}}{4} (x-3) = 1,93972 - 0,64675x$$

$$\langle u_3, \sin(x) \rangle = \int_1^5 \left(\frac{x^2 - 6x + \frac{23}{3}}{\sqrt{238514}} \right) \sin(x) dx = -0,125242 \rightarrow (-0,125242) \left(\frac{x^2 - 6x + \frac{23}{3}}{\sqrt{238514}} \right) = -0,0525093x^2 + 0,310056x - 0,402571$$

$$P(x) \approx 0,06416 + 1,93972 - 0,64675x - 0,0525093x^2 + 0,310056x - 0,402571$$

$$P(x) \approx -0,0525x^2 - 0,3316x + 1,6013$$

c) norma rozdílu $\|\sin(x) - (ax^2 + bx + c)\| = \sqrt{\int_1^5 (\sin(x) - (ax^2 + bx + c))^2 dx} = 0,319$